

Human Activity as a Driver of Mosquito Abundance in the Sierra Nevadas

Sylvia Li, Stephanie Copeland, Matthew Rosen, Hillary S. Young



Introduction

While mosquitoes can be vastly abundant in the Eastern Sierra Nevadas during the summer, populations of native vertebrate hosts are relatively low. For example, mammals such as Mule Deer (*Odocoileus hemionus*) have an estimated population being ~17,000 (as of 2017) [1]. However, the Eastern Sierras are a popular destination for hiking, camping and fishing which have made them a site for increased human recreational activity; an estimated four million people visit Inyo National Forest for recreational opportunities, heavily outnumbering deer as hosts [2]. The increase in human activity in the Eastern Sierras may play a part in driving mosquito populations, since host presence dictates mosquito reproduction and population success. This is seen in locations where a mosquito host is removed and mosquito populations decline or disappear in response [3].

Objective and Hypothesis

- Objective: Investigate the possible interactions between human recreational activity and mosquito populations in otherwise host-limited ecosystems.
- Research Questions: What are the primary hosts for mosquitos in alpine recreational areas of the Sierra Nevadas?
- Hypothesis: In high alpine recreational areas with low estimates of native vertebrate host populations, we expect that humans will be the predominant host for mosquitoes.

Methods

Aspirator traps were used to collect mosquito samples at various high alpine lake sites. Each lake was sampled at different times of day to account for mosquito population variation with time of day. Mosquitos were counted and gravid ones were separated. Gravid mosquito blood meals were collected on blood cards.

- 64 Bloodmeal samples were collected from 12,937 mosquitos, of which 47 yielded identifiable DNA.
- Genetic analysis was conducted:



Figure 1: Stephanie Copeland using an aspirator trap to collect samples from foliage surrounding lake sites.

DNA
Extraction

PCR 1

PCR 2

MySeq

NCBI
Blast [4]
via Galaxy

Field Work

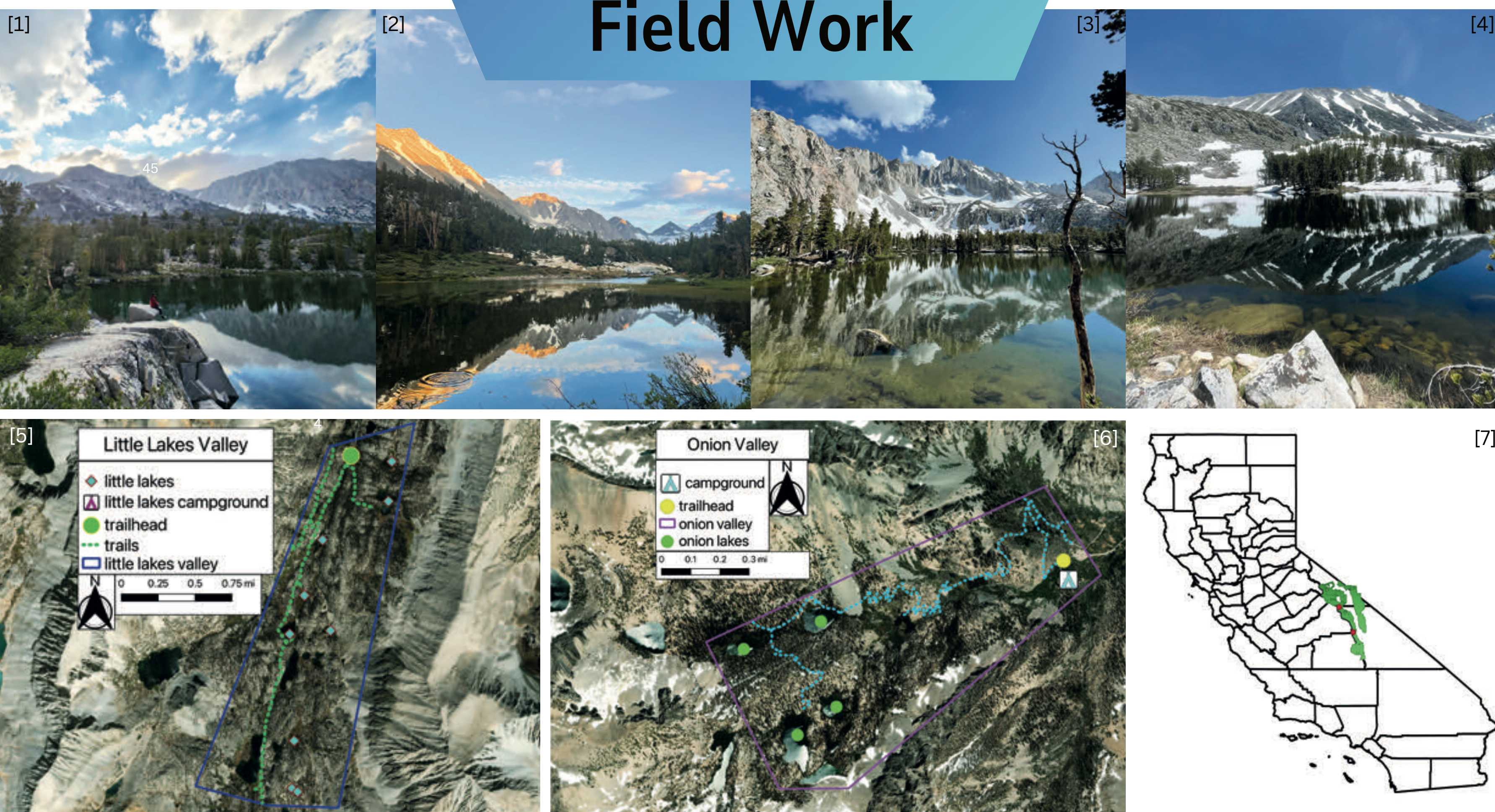


Figure 2: Collage of lake sampling sites and maps of site locations. [1] Little Fish Lake [2] Mack Lake [3] Matlock Lack [4] Hidden Lake [5] Map of Little Lakes Valley and relative lake locations [6] Map of Onion Valley and relative lake locations [7] Map of sampling sites compared to California

Results

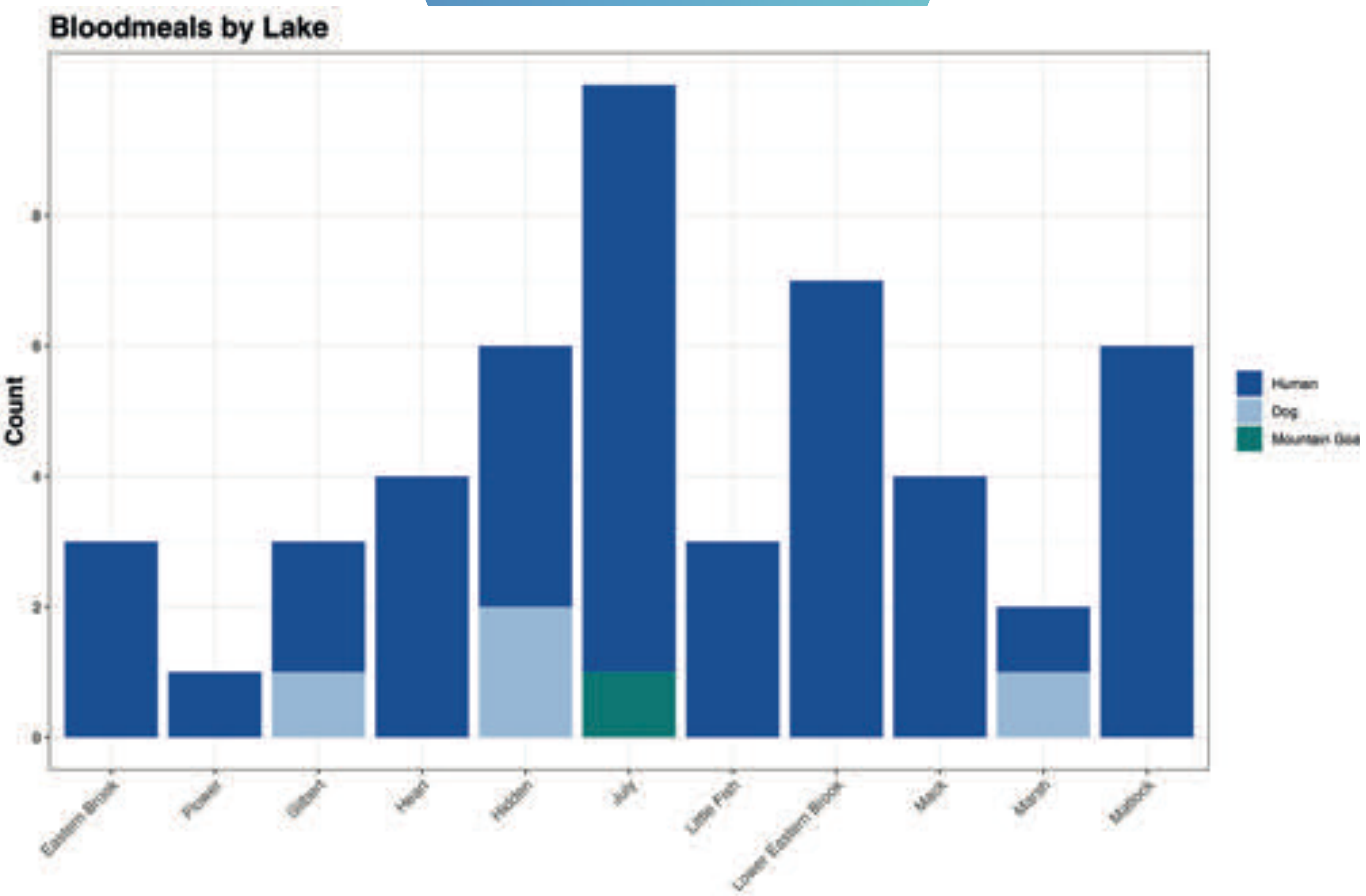


Figure 4: Distribution of bloodmeals according to the lake where the sample was taken from. A majority of samples contained human DNA, with a few samples containing canine DNA. One sample from 4th of July lake contained mountain goat DNA, an example of a native vertebrate host for mosquitos in the Eastern Sierras.

Results

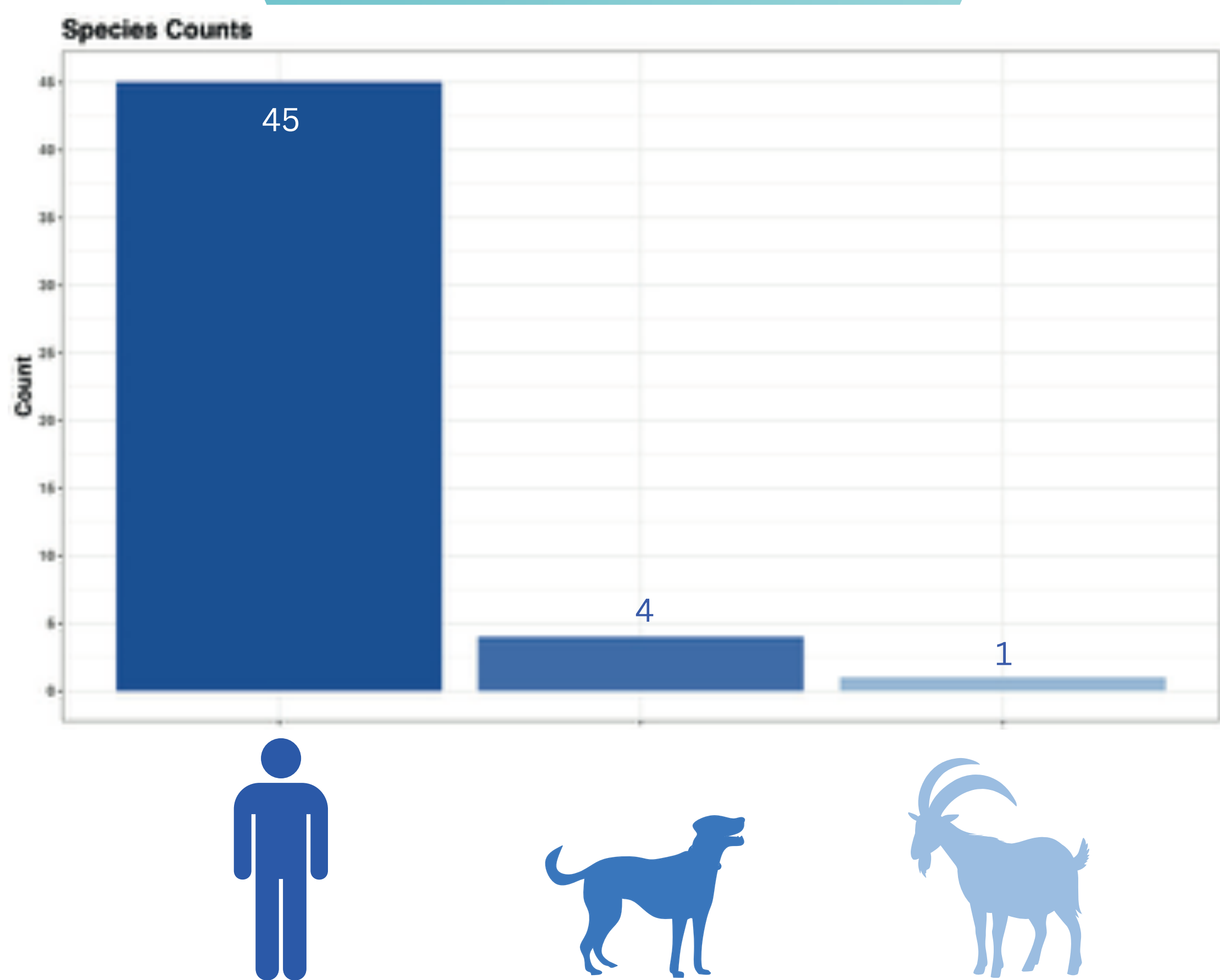


Figure 3: Counts for species with a high NCBI Blast percentage identity match for each bloodmeal sample. There was also evidence of multiple feedings, with a few samples containing DNA from two species.

Conclusions + Future Direction

Many of the bloodmeal samples contained human DNA, providing evidence that humans are a primary host species for mosquitos in the Eastern Sierras. This study sampled 12 lake sites at two highly visited recreational sites. In the next steps of this project, sampling could be conducted at more remote sites to see if results remain consistent. With more study and seasonal replication, bloodmeal analysis could act as a wider indicator of environmental health as human activity and interaction with the outdoors continue to increase.

References and Acknowledgements

[1] CA. Dept of Fish & Wildlife (2017) wildlife.ca.gov; [2] USFS (n.d.) Inyo NF History & Culture; [3] Lafferty, K.D. et al. (2018) *Biol. Letters*; [4] Verma et al. (2006) US Patent 7141364 B1

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Contact: sylvia_li@ucsb.edu

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